ASYMPTOTIC COUNTS OF GOLDBACH DECOMPOSITIONS IN THE NONNEGATIVE POLYNOMIAL SEMIRING

SHUNYAO YAN

ABSTRACT. For fixed integers $d \geq 1$ and $k \geq 2$, we count ordered representations of a polynomial $f = \sum_{i=0}^{d} c_i x^i \in \mathbb{N}_0[x]$ as a sum of exactly k multiplicative atoms of $\mathbb{N}_0[x]$. As the smallest coefficient tends to infinity, we prove an asymptotic formula whose main term is the total number of coefficientwise decompositions multiplied by an explicit, positive Euler product. The product depends on the target only through the prime divisors of its content. The error is uniform in the relative sizes of the coefficients. The proof combines residue counting in weak compositions, a large-prime estimate controlled by the shortest coefficient row, and a finite-field comparison of primitive and atomic summands. We also obtain a content-independent positive lower bound for the singular series, an explicit formula for two summands, and the average value $\zeta(d+1)^{-k}$ of the normalized representation count over expanding coefficient boxes.

1. Introduction and main result

Goldbach-type questions for polynomial rings concern representations as sums of multiplicatively irreducible polynomials; see, for example, Pollack [6]. In a semiring of nonnegative polynomials, the absence of subtraction changes both the multiplicative structure and the available additive representations. Gotti and Polo [2] studied factorization in polynomial semidomains. Liao and Polo [4] proved a two-atom Goldbach theorem for nonmonomial Laurent polynomials with coefficient sum greater than three, and Kaplan and Polo [3] extended this direction to more general Laurent polynomial and series semidomains. In an earlier unpublished work, Mittal, Pieleanu, Yan studied the asymptotic proportion of two-atom Goldbach decompositions in $\mathbb{N}_0[x]$. Building on this work, we generalize the analysis to ordered decompositions into exactly k multiplicative atoms for every fixed integer $k \geq 2$, obtaining an explicit asymptotic formula.

Here we study an enumeration problem in the ordinary semiring $\mathbb{N}_0[x]$. We fix both the degree of the target and the number of summands, and allow all target coefficients to grow, without requiring comparable rates of growth. The main term records the congruence dependence between summands whose sum is prescribed. In particular, the individual densities of primitive coefficient vectors cannot simply be multiplied: the content of the target remains visible in the answer. The familiar zeta factors in primitive-vector density [5] reappear after averaging over the target.

Throughout, $\mathbb{N} = \{1, 2, \ldots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. An *atom* of $\mathbb{N}_0[x]$ is a nonzero nonunit h such that $h = uv$, with $u, v \in \mathbb{N}_0[x]$, implies that one of u, v is a unit. The only unit is 1; write $\mathcal{A}$ for the set of atoms. Atomicity in this paper is always multiplicative atomicity in $\mathbb{N}_0[x]$, not irreducibility over $\mathbb{Q}$. For example, $1 + x^3$ is an atom of $\mathbb{N}_0[x]$, although it factors in $\mathbb{Q}[x]$.

For $k \geq 2$, define

$$G_k(f) = \#\{(g_1, \ldots, g_k) \in \mathcal{A}^k : g_1 + \cdots + g_k = f\}.$$

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The summands are ordered, repetitions are allowed, and their degrees need not equal the degree of f. Prime constants and the atom x are included. For

$$f(x) = \sum_{i=0}^{d} c_i x^i, \qquad c_i \in \mathbb{N}, \tag{1}$$

put

$$n = d + 1, \qquad C = \gcd(c_0, \dots, c_d), \qquad M = \min_{0 \le i \le d} (c_i + 1). \tag{2}$$

The number of all ordered coefficientwise decompositions, without any atomicity condition, is

$$V_k(f) = \prod_{i=0}^{d} \binom{c_i + k - 1}{k - 1}. \tag{3}$$

Here the notation V_k denotes a count, not a continuous volume.

For a prime p, set

$$\rho_{p,n,k}(C) = (1 - p^{-n})^k + (-1)^k (p^n \mathbf{1}_{p|C} - 1) p^{-nk}, \tag{4}$$

and define the singular series

$$\mathfrak{S}_{n,k}(C) = \prod_{p} \rho_{p,n,k}(C). \tag{5}$$

Theorem 1.1 (Uniform fixed-degree, fixed-length asymptotic)**.** *Fix $d \ge 1$ and $k \ge 2$, and let f satisfy (1). The product (5) converges absolutely and satisfies*

$$\frac{3}{4\zeta(d+1)^k} \le \mathfrak{S}_{d+1,k}(C) \le 1. \tag{6}$$

As $M \to \infty$,

$$G_k(f) = \big(\mathfrak{S}_{d+1,k}(C) + o_{d,k}(1)\big) \prod_{i=0}^{d} \binom{c_i + k - 1}{k - 1}. \tag{7}$$

The error is uniform over all positive coefficient vectors: for every $\varepsilon > 0$ there exists $M_0(d, k, \varepsilon)$ such that

$$\left| \frac{G_k(f)}{V_k(f)} - \mathfrak{S}_{d+1,k}(C) \right| \le \varepsilon \quad \text{whenever } M \ge M_0(d, k, \varepsilon).$$

Consequently, (7) is also a uniform relative asymptotic, and every target with sufficiently large M is a sum of exactly k atoms.

The parameters d and k are fixed in this theorem. No bound on $\max_i c_i / \min_i c_i$ is imposed. In particular, the content C may vary with the target and need not tend to a limit. We make no claim here for growing d or k, or for targets with fixed zero coefficients.

The proof has two parts. First we count decompositions with primitive coefficient columns and derive their local factors by inclusion–exclusion. The large-prime tail is controlled using the shortest row of coefficients. Second, we show that primitive and atomic columns disagree with probability tending uniformly to zero. For that step, we prove the needed irreducibility-density estimate directly for the nonuniform column distributions arising from decompositions.

2. Coefficient distributions and residues

For a nonnegative integer c, let

$$\mathcal{C}_k(c) = \{(a_1, \dots, a_k) \in \mathbb{N}_0^k : a_1 + \cdots + a_k = c\}.$$

Choose a vector uniformly from $\mathcal{C}_k(c_i)$, independently for each $i = 0, \dots, d$. Writing its entries as $A_{i,1}, \dots, A_{i,k}$ and setting

$$g_j(x) = \sum_{i=0}^{d} A_{i,j} x^i$$

gives the uniform distribution on the $V_k(f)$ decompositions in (3). Rows are independent; columns generally are not.

The content of a nonzero integer polynomial is the nonnegative gcd of its coefficients, and we set $\operatorname{cont}(0) = 0$. A polynomial or coefficient column is *primitive* when its content is one. Let $P_k(f)$ count the decompositions whose k columns are all primitive. Thus $P_k(f)/V_k(f)$ and $G_k(f)/V_k(f)$ are probabilities in the preceding model.

For probability measures on a finite set, write

$$\mathrm{TV}(\mu, \nu) = \frac{1}{2} \sum_a |\mu(a) - \nu(a)|.$$

The discrepancy on any event is at most this quantity. Marginalization cannot increase it, and for product measures it is at most the sum of the discrepancies of the individual factors. The latter assertion follows by telescoping the difference of the two products.

Lemma 2.1 (Residues of a weak composition)**.** *Let $q \geq 2$, and choose a uniform vector from $\mathcal{C}_k(c)$. The compatible residue vectors modulo q are those whose coordinate sum is c modulo q, and there are q^{k-1} of them. If $c \geq 2kq$, the residue law has total-variation distance at most*

$$L_k \frac{q}{c}, \qquad L_k = \frac{k(k-1)}{2} \left(\frac{3}{2}\right)^{k-2}, \tag{8}$$

from the uniform law on these compatible vectors.

Proof. Fix representatives $r_1, \dots, r_k \in \{0, \dots, q-1\}$ with compatible sum, and write $t = \sum_j r_j$. The entries of a lift are $r_j + q b_j$, where

$$b_1 + \cdots + b_k = (c - t)/q.$$

The exact number of lifts is

$$\binom{(c-t)/q + k - 1}{k-1}, \tag{9}$$

interpreted as zero when $t > c$. Under $c \geq 2kq$ we have $t < c$. The ratio of the probability of this residue vector to q^{1-k} is

$$\prod_{h=1}^{k-1} \frac{c - t + qh}{c + h}.$$

Each factor differs from one in absolute value by at most $kq/c \leq 1/2$. Telescoping the product gives a difference from one of at most

$$(k-1) \left(\frac{3}{2}\right)^{k-2} \frac{kq}{c}.$$

Averaging this bound over the q^{k-1} residues and dividing by two proves (8). □

Set

$$A_{n,k} = n L_k. \tag{10}$$

Corollary 2.2 (Matrix and column residues)**.** *If $M-1 \geq 2kq$, the full coefficient matrix modulo q differs in total variation by at most*

$$A_{n,k}\frac{q}{M-1} \tag{11}$$

from the uniform law on matrices with the prescribed row sums modulo q. Under this comparison law, each designated column is uniform on $(\mathbb{Z}/q\mathbb{Z})^n$. The same bound therefore holds for its marginal law.

Proof. Apply Lemma 2.1 to each row and add the bounds. In a compatible row, fixing one entry leaves q^{k-2} choices for the remaining entries, including the case $k=2$. Thus that entry is uniform modulo q. Independence of the rows proves the column assertion. □

Lemma 2.3 (A decreasing entry distribution)**.** *A designated entry in a uniform vector from $\mathcal{C}_k(c)$ has mass function*

$$w_c(a) = \frac{\binom{c-a+k-2}{k-2}}{\binom{c+k-1}{k-1}}, \qquad 0 \leq a \leq c. \tag{12}$$

It is nonincreasing, and

$$w_c(0) = \frac{k-1}{c+k-1}.$$

For every positive integer p,

$$\mathbb{P}(p \mid A_{i,j}) \leq w_{c_i}(0) + \frac{1}{p}. \tag{13}$$

In particular, if

$$\beta_M = \frac{k-1}{M+k-2}, \tag{14}$$

then $w_{c_i}(0) \leq \beta_M$ for every row, and

$$\mathbb{P}(p \mid A_{i,j}) \leq \frac{k}{p} \quad \text{for } p \leq M-1. \tag{15}$$

Proof. Fixing the designated entry leaves a weak composition of $c-a$ into $k-1$ parts, which gives (12). The binomial coefficient in the numerator is nonincreasing in a.

For any nonincreasing probability mass w on $\mathbb{N}_0$, extended by zero outside its support, the disjoint blocks $\{(m-1)p+1, \ldots, mp\}$ give

$$p\sum_{m\geq 1} w(mp) \leq \sum_{a\geq 1} w(a) \leq 1.$$

Separate the zero entry to obtain (13). Finally, $\beta_M \leq (k-1)/(M-1) \leq (k-1)/p$ when $p \leq M-1$. □

3. The primitive count and its singular series

Lemma 3.1 (The local probability)**.** *Under uniform compatible residues modulo a prime p, the probability that none of the k columns is zero is $\rho_{p,n,k}(C)$ from (4).*

Proof. Put $Q = p^n$ and $\mathbf{c} = (c_0, \ldots, c_d) \bmod p$. There are Q^{k-1} tuples of vectors in $\mathbb{F}_p^n$ with sum $\mathbf{c}$. Requiring any specified $r < k$ columns to vanish has probability Q^{-r}. Requiring all k to vanish has probability $\mathbf{1}_{\mathbf{c}=0}Q^{1-k}$. Inclusion–exclusion therefore gives

$$\sum_{r=0}^{k-1}(-1)^r\binom{k}{r}Q^{-r} + (-1)^k\mathbf{1}_{\mathbf{c}=0}Q^{1-k} = (1-Q^{-1})^k + (-1)^k(Q\mathbf{1}_{\mathbf{c}=0}-1)Q^{-k}.$$

The condition $\mathbf{c}=0$ is exactly $p \mid C$. □

For an integer $y \geq 2$, write

$$q_y = \prod_{p \leq y} p, \qquad T_n(y) = \sum_{m > y} m^{-n} \leq \frac{y^{1-n}}{n-1}. \tag{16}$$

The final inequality uses $n \geq 2$.

Proposition 3.2 (Convergence and uniform positivity)**.** *For all $n, k \geq 2$ and $C \geq 1$, the series (5) converges absolutely, satisfies (6), and obeys*

$$0 \leq \prod_{p \leq y} \rho_{p,n,k}(C) - \mathfrak{S}_{n,k}(C) \leq k T_n(y). \tag{17}$$

Proof. Each local factor is a probability. A union bound over the k zero-column events gives

$$0 \leq 1 - \rho_{p,n,k}(C) \leq k p^{-n}.$$

The sum of the right side converges. Telescoping finite products of numbers in $[0, 1]$, followed by passage to the limit, proves (17).

For the positive lower bound, write

$$\rho_{p,n,k}(C) = (1 - p^{-n})^k (1 + e_p), \qquad e_p = \frac{(-1)^k (p^n \mathbf{1}_{p|C} - 1)}{(p^n - 1)^k}.$$

If k is even, the only negative case is $p \nmid C$, with $|e_p| = (p^n - 1)^{-k}$. If k is odd, then $k \geq 3$ and the only negative case is $p \mid C$, with $|e_p| = (p^n - 1)^{1-k}$. In either case, the negative part $u_p = \max(0, -e_p)$ satisfies

$$u_p \leq (p^n - 1)^{-2}.$$

Consequently,

$$\sum_p u_p \leq \sum_p \frac{1}{(p^2 - 1)^2} \leq \frac{1}{3} \sum_{m=2}^{\infty} \frac{1}{m^2 - 1} = \frac{1}{4}. \tag{18}$$

For the last equality, use $1/(m^2 - 1) = \frac{1}{2}((m-1)^{-1} - (m+1)^{-1})$.

For every finite prime set, $\prod_p (1 + e_p) \geq \prod_p (1 - u_p) \geq 1 - \sum_p u_p \geq 3/4$. The Euler product for $\zeta(n)$ now gives $\mathfrak{S}_{n,k}(C) \geq 3/(4\zeta(n)^k)$. This also proves that no local factor or limiting product vanishes. □

Lemma 3.3 (Uniform removal of large common primes)**.** *The probability that some column has a common prime divisor greater than y is at most*

$$k\beta_M + k^{n+1} T_n(y). \tag{19}$$

Here a zero column is regarded as divisible by every prime.

Proof. Choose a row of minimum total $M - 1$. For one designated column, its entry in that row is zero with probability at most β_M. Otherwise any prime dividing the entire column is at most $M - 1$.

For a prime p in that range, the entries of the designated column are independent across rows. By Lemma 2.3, the probability that they are all divisible by p is at most $(k/p)^n$. Summing over $y < p \leq M - 1$ and then over the k columns gives (19). The bound contains no largest-coefficient term. □

Proposition 3.4 (Primitive decompositions)**.** *If $M - 1 \geq 2kq_y$, then*

$$\left| \frac{P_k(f)}{V_k(f)} - \mathfrak{S}_{n,k}(C) \right| \leq A_{n,k} \frac{q_y}{M-1} + k\beta_M + (k^{n+1} + k) T_n(y). \tag{20}$$

Proof. Let $\mathcal{E}_y$ be the event that no column is zero modulo any prime at most y. The Chinese remainder theorem identifies the uniform compatible law modulo q_y with the product of its compatible laws at those primes. Lemma 3.1 and Corollary 2.2 give

$$\left|\mathbb{P}(\mathcal{E}_y) - \prod_{p\le y} \rho_{p,n,k}(C)\right| \le A_{n,k}\frac{q_y}{M-1}.$$

Primitivity of every column implies $\mathcal{E}_y$. Conversely, a column that passes this finite sieve but is not primitive has a common prime greater than y. Zero columns are also covered by Lemma 3.3. Apply that lemma and then (17). □

4. From primitive columns to atomic summands

Lemma 4.1 (Atomicity and content)**.** *The constant atoms of $\mathbb{N}_0[x]$ are the positive integer primes, and the only atom with zero constant coefficient is x. Every nonconstant atom is primitive. A primitive nonconstant polynomial that is irreducible over $\mathbb{Q}$ is an atom of $\mathbb{N}_0[x]$.*

Proof. Degree comparison shows that an invertible polynomial must be constant, and the only unit in $\mathbb{N}_0$ is 1. A factorization of a positive constant likewise has constant factors, proving the prime assertion. A nonzero zero-constant polynomial is xh; this is atomic only when $h = 1$, and x is indeed an atom.

If a nonconstant polynomial has content $c > 1$, factoring out c is a product of two nonunits. Thus a nonconstant atom is primitive. For a primitive nonconstant polynomial, a constant factor in $\mathbb{N}_0[x]$ must be 1. Any nontrivial factorization would therefore have two positive-degree factors, contradicting rational irreducibility. □

For instance, $2 + 2x = 2(1 + x)$ is irreducible as a degree-one polynomial over $\mathbb{Q}$, but is not a semiring atom. Conversely, $1 + x^3$ is an atom: in a nonnegative factorization its positive factor exponents would all have to occur in $\{0, 3\}$, which precludes two positive-degree factors. These examples explain why primitivity is needed in the comparison.

The following elementary count provides the only finite-field irreducibility input. We include its proof; the standard finite-field construction and subfield facts are described in Conrad [1].

Lemma 4.2 (A positive density of modular certificates)**.** *Fix $d \ge 2$. If $p \ge 4d^2$ is prime, the proportion of vectors in $\mathbb{F}_p^{d+1}$ representing an irreducible polynomial of degree exactly d is at least*

$$\alpha_d = \frac{1}{4d}. \tag{21}$$

Proof. In a splitting field over $\mathbb{F}_p$, the roots of $X^{p^d} - X$ form a field: Frobenius gives closure under addition and multiplication, and a nonzero root has its inverse in the same set. The derivative is -1, so this field has exactly p^d elements.

An element's degree r over $\mathbb{F}_p$ divides d. If $r < d$, it is a root of $X^{p^r} - X$, and $r \le d/2$. Thus at most

$$\sum_{\substack{r\mid d\\ r<d}} p^r \le dp^{d/2} \le \tfrac{1}{2}p^d$$

elements have degree smaller than d, where the final inequality holds for the stated range of p.

Every monic irreducible of degree d accounts for at most d elements. There are therefore at least $p^d/(2d)$ such polynomials. Their $p - 1$ nonzero scalar multiples are all distinct, so their proportion among p^{d+1} coefficient vectors is at least

$$\frac{(p-1)p^d}{2d\,p^{d+1}} = \frac{1 - 1/p}{2d} \ge \frac{1}{4d}.$$

□

Lemma 4.3 (Rational reducibility under the column law)**.** *Fix $d \geq 2$, and choose distinct primes $p_1, \ldots, p_\ell \geq 4d^2$. Put*

$$R_\ell = p_1 \cdots p_\ell.$$

If $M - 1 \geq 2kR_\ell$, then a designated summand column satisfies

$$\mathbb{P}(\deg g_j = d,\ g_j \text{ reducible over } \mathbb{Q}) \leq (1 - \alpha_d)^\ell + A_{n,k}\frac{R_\ell}{M-1}. \tag{22}$$

Proof. A rationally reducible integer polynomial of degree d has, by Gauss's lemma, a factorization into two positive-degree integer polynomials. If its reduction has degree d, the leading coefficients of both factors remain nonzero, so both factor degrees are preserved. Its reduction cannot then be an irreducible of degree d.

Under uniform column residues modulo R_ℓ, CRT makes the reductions at the ℓ primes independent. By Lemma 4.2, the probability that all ℓ degree-preserving irreducibility certificates fail is at most $(1 - \alpha_d)^\ell$. Transfer this event bound to the actual column law using Corollary 2.2. □

Remark 4.4 (Degree preservation is essential)*.* The polynomial

$$(1 + 2x)(1 + x) = 1 + 3x + 2x^2$$

is rationally reducible, but reduces modulo two to the irreducible linear polynomial $1 + x$. Lemma 4.3 uses irreducible reductions of degree *exactly* d and therefore does not make this erroneous inference.

Proposition 4.5 (Uniform primitive-to-atom comparison)**.** *For $d \geq 2$ and the choices in Lemma 4.3,*

$$\left|\frac{G_k(f) - P_k(f)}{V_k(f)}\right| \leq 2k\beta_M + k(1 - \alpha_d)^\ell + kA_{n,k}\frac{R_\ell}{M-1}. \tag{23}$$

For $d = 1$, the left side is at most $k\beta_M$. Consequently, for every fixed $d \geq 1$ and $k \geq 2$,

$$\frac{G_k(f) - P_k(f)}{V_k(f)} = o_{d,k}(1) \tag{24}$$

uniformly as $M \to \infty$.

Proof. For one column, discard the events that its constant or degree-d coefficient is zero. Their combined probability is at most $2\beta_M$ by Lemma 2.3. Outside this boundary, the column has positive constant coefficient and degree d.

By Lemma 4.1, every atomic such column is primitive; if it is primitive but not atomic, it is rationally reducible. Hence the probability that its primitive and atomic indicators differ is bounded by $2\beta_M$ plus (22). A union bound over columns bounds the symmetric difference between the events "all primitive" and "all atomic," proving (23). No independence between columns is used.

When $d = 1$, every nonconstant column is atomic exactly when primitive: a linear factorization can have only one positive-degree factor, so the other factor divides the content. This includes x. Only constant columns can cause a discrepancy, and each has probability at most β_M because its linear coefficient is zero.

For $d \geq 2$, first fix ℓ large enough to make the exponential term small, and then let M grow. The primes, and hence R_ℓ, stay fixed in this limit. This proves the asserted uniformity. □

Remark 4.6 (The two counting events need not contain one another)*.* Prime constants are atoms but not primitive, whereas the constant polynomial 1 is primitive but not an atom. Accordingly, one cannot replace the symmetric-difference argument above by an unconditional inclusion between the two sets of decompositions. For example, direct linear counting gives

f	$G_2(f)$	$P_2(f)$	$V_2(f)$
$2 + 30x$	31	33	93
$30 + 2x$	49	33	93

These finite boundary effects are included in the error term.

5. Completion of the asymptotic formula

The preceding estimates make the order of limits explicit. With the notation already defined, if $d \geq 2$ and

$$M - 1 \geq 2k \max(q_y, R_\ell),$$

Propositions 3.4 and 4.5 give

$$\left| \frac{G_k(f)}{V_k(f)} - \mathfrak{S}_{n,k}(C) \right| \leq 3k\beta_M + (k^{n+1} + k)T_n(y) + k\left(1 - \frac{1}{4d}\right)^{\ell} + A_{n,k}\frac{q_y + kR_\ell}{M-1}. \tag{25}$$

For $d = 1$, the sharper bound is

$$\left| \frac{G_k(f)}{V_k(f)} - \mathfrak{S}_{2,k}(C) \right| \leq 2k\beta_M + (k^3 + k)T_2(y) + A_{2,k}\frac{q_y}{M-1}, \tag{26}$$

provided $M - 1 \geq 2kq_y$.

Proof of Theorem 1.1. Convergence and (6) were proved in Proposition 3.2. Fix d, k and $\varepsilon > 0$. First choose the prime cutoff y so that the $T_n(y)$ term in (25) is small. For $d \geq 2$, next choose ℓ and then a fixed set of ℓ sufficiently large distinct primes so that the exponential term is small. Finally choose M large enough for the displayed size hypotheses and for all remaining terms to be small. None of these choices depends on the largest coefficient, on the aspect ratios, or on the target content. For $d = 1$, use (26) without the second prime set.

This proves the uniform absolute asymptotic (7). Since $\mathfrak{S}_{n,k}(C) \geq 3/(4\zeta(n)^k)$ for every content, division by the main-term factor gives the uniform relative asymptotic. Taking the absolute error below half of this lower bound gives

$$\frac{G_k(f)}{V_k(f)} \geq \frac{3}{8\zeta(n)^k} > 0$$

for all sufficiently large M, proving existence of k-atom representations. □

The use of fixed prime cutoffs is deliberate. Finite-modulus equidistribution alone does not justify allowing the modulus to grow arbitrarily with the coefficient lengths. Here the large-prime error is removed separately, through the shortest-row estimate, before any claim of uniform asymptotics is made.

6. Content dependence and averaging

Corollary 6.1 (Two atomic summands)**.** *For fixed $d \geq 1$ and targets with all coefficients tending to infinity,*

$$G_2(f) = \left(\prod_{p \mid C} (1 - p^{-(d+1)}) \prod_{p \nmid C} (1 - 2p^{-(d+1)}) + o_d(1) \right) \prod_{i=0}^{d} (c_i + 1), \tag{27}$$

uniformly in the relative coefficient sizes.

Proof. For $k = 2$, formula (4) equals $1 - p^{-n}$ when $p \mid C$ and $1 - 2p^{-n}$ otherwise. Apply Theorem 1.1. □

The singular series depends only on the prime divisors of C, not on their exponents. Its dependence on a newly introduced content prime also changes with the parity of k:

$$\rho_{p,n,k}(p \mid C) - \rho_{p,n,k}(p \nmid C) = (-1)^k p^{-n(k-1)}. \tag{28}$$

For instance, with $u = p^{-n}$, the three-summand local factors are $1 - 3u + 2u^2$ when $p \mid C$ and $1 - 3u + 3u^2$ otherwise.

Example 6.2 (Two different target limits)**.** Fix $d \geq 1$ and put $n = d+1$. For

$$f_H(x) = \sum_{i=0}^{d} (H+i)x^i,$$

the content is one, because the first two coefficients are consecutive. Hence

$$\frac{G_2(f_H)}{V_2(f_H)} \longrightarrow \prod_p (1-2p^{-n}).$$

For the targets

$$h_H(x) = H!\,(1 + x + \cdots + x^d),$$

every prime at most H divides the content. Formula (27) and the uniform product-tail estimate give

$$\frac{G_2(h_H)}{V_2(h_H)} \longrightarrow \prod_p (1-p^{-n}) = \zeta(n)^{-1}.$$

These are distinct limits, since each factor $1-2p^{-n}$ is strictly smaller than $1-p^{-n}$. Thus the normalized count has no single content-independent limit for arbitrary target sequences.

Averaging the target restores a simpler value. The next result averages the *normalized count for each target*; it does not replace this average by a ratio of two separately summed counts.

Theorem 6.3 (Average over target boxes)**.** *Fix $d \geq 1$, $k \geq 2$, and let c_i be independent and uniform on $\{1, \ldots, H_i\}$. Put $f = \sum_i c_i x^i$ and $C = \gcd_i c_i$. As $\min_i H_i \to \infty$,*

$$\mathbb{E}\,\mathfrak{S}_{d+1,k}(C) \longrightarrow \zeta(d+1)^{-k}, \qquad \mathbb{E}\,\frac{G_k(f)}{V_k(f)} \longrightarrow \zeta(d+1)^{-k}. \tag{29}$$

No restriction on the ratios of the H_i is required.

Proof. Fix $y \geq 2$. As every H_i grows, the target coefficient vector becomes uniform modulo q_y. CRT makes its reductions at the primes $p \leq y$ independent in the limiting law. At a single prime,

$$\mathbb{P}(p \mid C) \longrightarrow p^{-n}.$$

Taking the expectation of (4), the correction term vanishes:

$$\mathbb{E}\,\rho_{p,n,k}(C) \longrightarrow (1-p^{-n})^k,$$

because $p^n \cdot p^{-n} - 1 = 0$. Independence in the limiting finite residue law therefore gives

$$\mathbb{E} \prod_{p \leq y} \rho_{p,n,k}(C) \longrightarrow \prod_{p \leq y} (1-p^{-n})^k.$$

The uniform bound (17) permits $y \to \infty$, proving the first limit in (29).

For the second, fix $\varepsilon > 0$ and take the threshold M_0 in Theorem 1.1. The probability that $\min_i(c_i+1) < M_0$ is at most

$$\sum_{i=0}^{d} \frac{M_0 - 1}{H_i},$$

which tends to zero. Outside this event the difference between the normalized count and its singular series is at most ε; on the event it is at most one, since both quantities lie in $[0,1]$. Their expectations therefore differ by a quantity tending to zero. □

The main term thus separates two effects: the product of binomial coefficients counts nonnegative coefficient allocations, while the Euler product records simultaneous primitivity under the prescribed sum. Rational reducibility and exceptional constant or degree-drop summands contribute only to the vanishing error in the fixed-d, fixed-k limit. This gives both a uniform count for each target and an arithmetic explanation of the simpler average density.

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Email address: shunyaoyan2009@gmail.com